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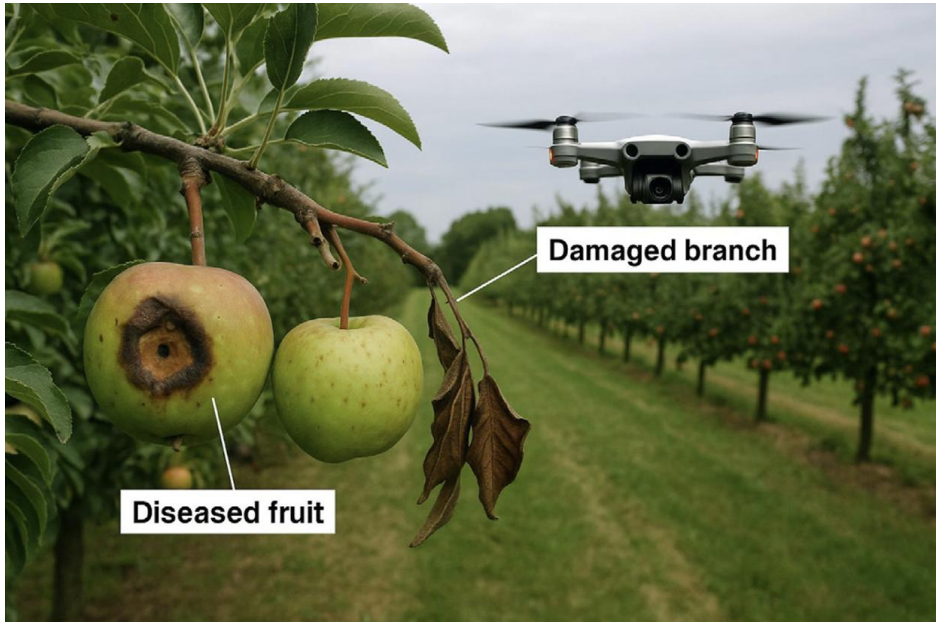
Agentic AI Infrastructure

Master Informatik
Seminar
Rico Groß



AI Agents

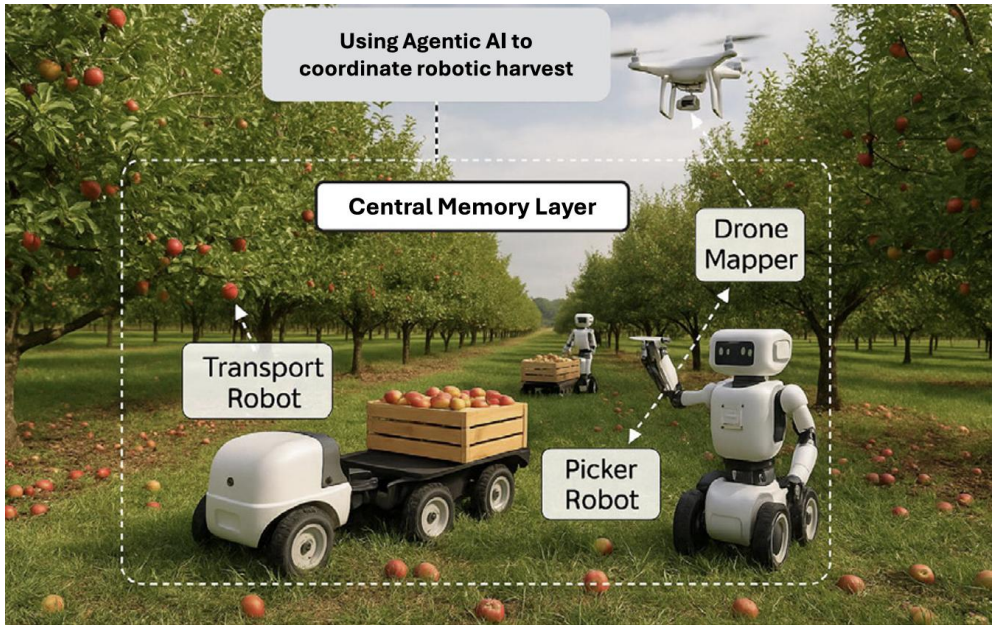
- Autonomous software systems to fulfill goal-directed tasks within bounded environments
- Build upon LLMs as reasoning engines
- Utilize external tools and data sources
- Key characteristics:
  - Autonomy: operate with minimal or no human intervention
  - Task-specificity: execute precisely specified tasks within a narrow domain (e.g. email filtering)
  - Reactivity and Adaptation: perceive and interact with dynamic inputs from real-time environments
- AI Agents transform static LLMs into a system capable of perceiving, planning, and acting



R. Sapkota, K. I. Roumeliotis, and M. Karkee, “Ai agents vs. agentic ai: A conceptual taxonomy, applications and challenges,” Information Fusion, vol. 126, p. 103599.

Agentic AI

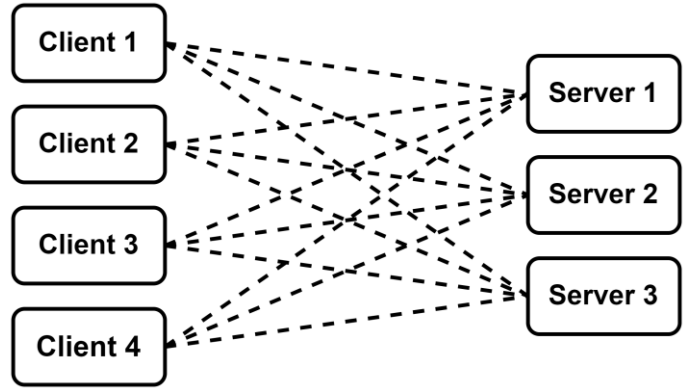
- Single-agent architecture falls short in more advanced use cases → development of a new paradigm: Agentic AI
- Extends capabilities of single AI Agents by enabling multiple AI Agents to work together through
  - Structured communication
  - Shared memory
  - Dynamic role assignment
- Multiple specialized AI Agents collaborate to achieve complex, high-level objectives
- Introduces system-level intelligence characterized by dynamic inter-agent collaboration
- Typically exceeds the operational limits of individual AI Agents



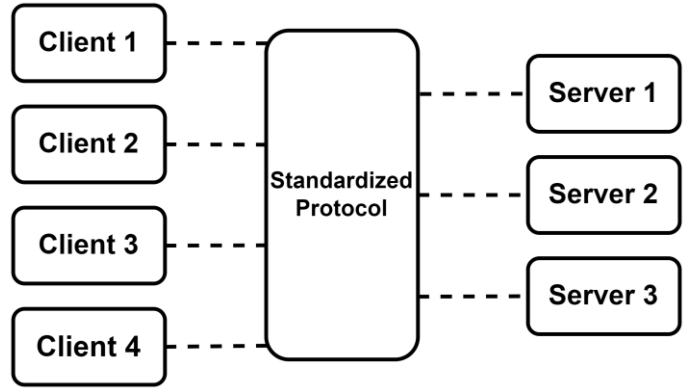
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The Scaling Bottleneck - MxN Integration Problem

- Significant challenge in the infrastructure for AI ecosystems
- Occurs when multiple clients connect to multiple servers
- Without a standardized method, each client is required to implement a custom integration to connect to each server → M x N integrations necessary

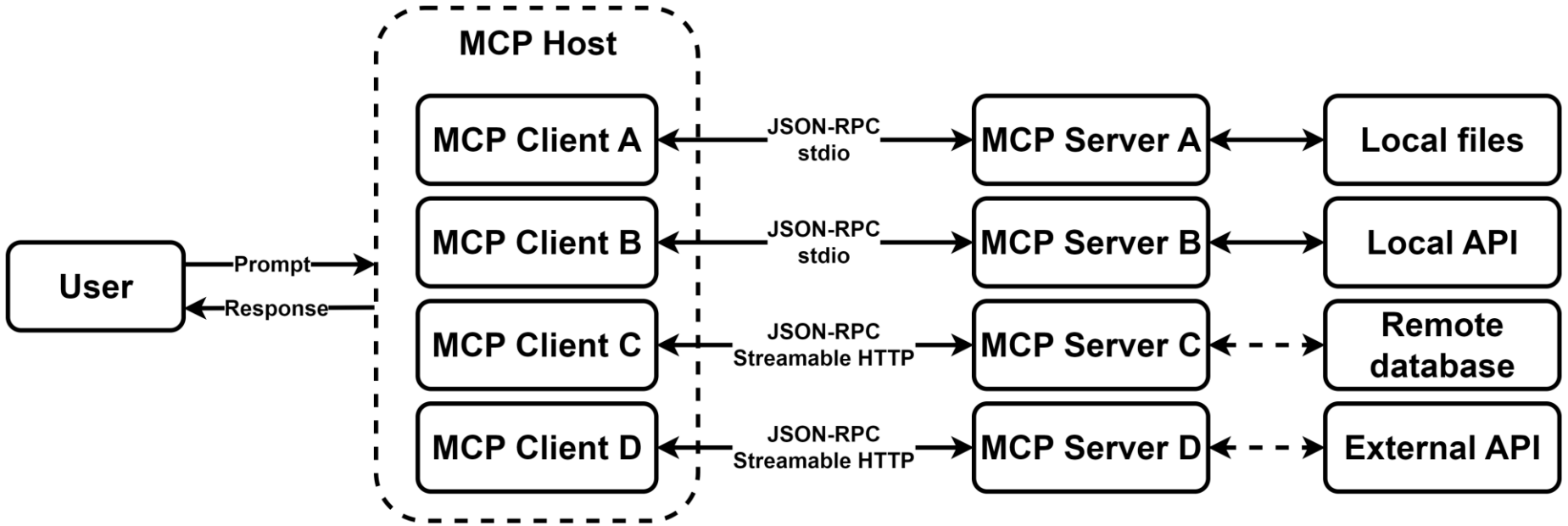


- In Agentic AI systems, the problem can occur across three layers of interaction
- Three standardized protocols emerged to solve the MxN integration problem → only M + N integrations of the standardized protocol



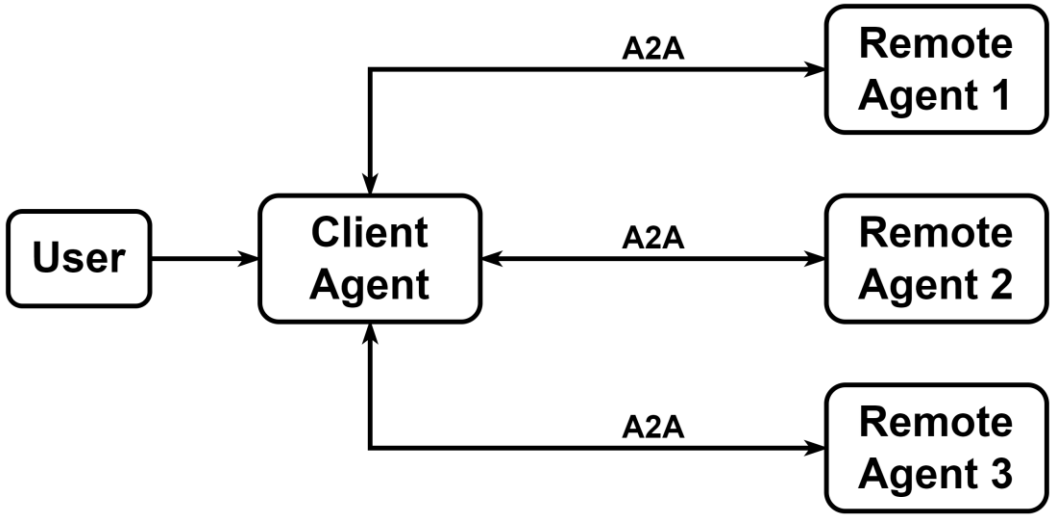
Model Context Protocol (MCP)

- Standardizes how AI Agents connect to existing data sources
- Data sources run a MCP Server to expose their capabilities to AI Agents
- The Network Effect: Open architecture → value of MCP scales with its adoption across models and servers
- Servers expose Tools, Resources and Prompts
- Clients provide Roots, Elicitation and Sampling



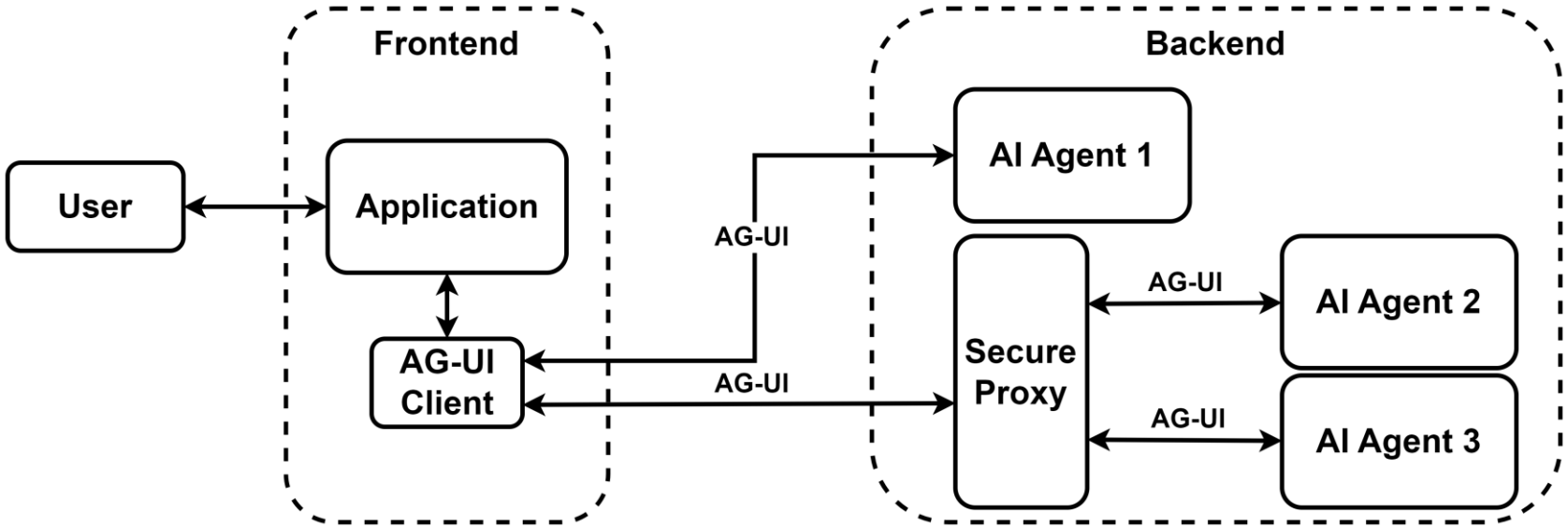
Agent-to-Agent (A2A) Protocol

- Standardizes how multiple AI Agents collaboratively work together
- Treats AI Agents as opaque entities → they collaborate without exposing internal logic, prompts or proprietary architectures
- Workflow:
  - User initiates a request
  - Discovery of AI Agents through their Agent Cards
    - Agent Card: JSON metadata file acting as digital business card detailing skills, endpoints and capabilities
  - Authentication handshake between Client Agent and chosen Remote Agent
  - Task Execution of client-submitted tasks through Remote Agent



Agent-User Interaction (AG-UI) Protocol

- Standardizes how AI Agents connect to frontend applications
- Decouples backend logic from frontend display
- Streams structured events, intents and state changes
- Adds transparency to the “AI-Blackbox”
  - Users can observe the AI “think”
- Transport Agnostic
- Enables real-time UI updates and transparent collaborative execution



How MCP, A2A and AG-UI build the backbone of Agentic AI

